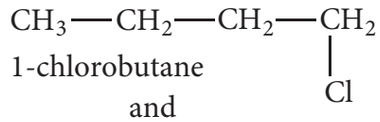
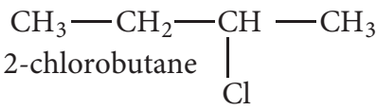
(ii) Mol. formula C_4H_9Cl



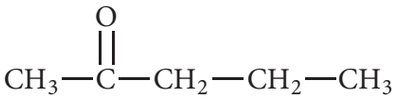
1-chlorobutane

and



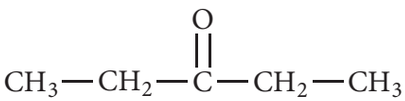
2-chlorobutane

(iii) Mol. formula $C_5H_{10}O$



Pent-2-one

and



Pent-3-one

(c) Functional isomerism:

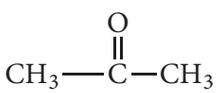
Different compounds having same molecular formula but different functional groups are said to exhibit functional isomerism.

Example:

(i) C_3H_6O $CH_3—CH_2·CHO$

Propanal

(aldehyde group)



Propanone

(keto group)

(ii) $C_3H_6O_2$ $CH_3—CH_2—COOH$

Propanoic acid

acid group

$CH_3—COOCH_3$

Methyl acetate

(ester group)

(d) Metamerism: This type of isomerism is a special kind of structural isomerism arises due to the unequal distribution of carbon atoms on either side of the functional group or different alkyl groups attached to the either side of the same functional group and having same molecular formula. This isomerism is shown by compounds having functional group such as ethers, ketones, esters and secondary amines between two alkyl groups.

(i) $C_4H_{10}O$ $CH_3—O—C_3H_7$

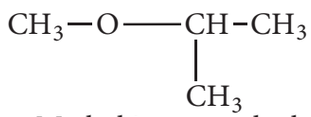
Methyl propyl ether

1-methoxypropane

$C_2H_5—O—C_2H_5$

diethyl ether

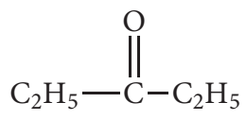
ethoxyethane



Methyl iso-propyl ether

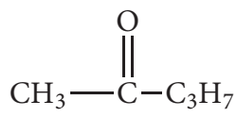
2-methoxypropane

(ii) $C_5H_{10}O$



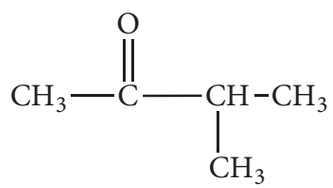
diethyl ketone

pentan-3-one



Methyl propyl ketone

pentan-2-one



Methyl isopropyl ketone

3-methylbutan-2-one

Unit 11 revised - jagan.indd 133 22-01-2021 12:42:26